



EVOLUTION OF LIFE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Geological Epochs — Oldest to Newest

- Source chronology oldest se newest: Paleocene → Eocene → Oligocene → Miocene → Pliocene → Pleistocene → Holocene. Given options mein correct sequence Eocene → Miocene → Pliocene → Pleistocene → Holocene hai. [Q1]

2. Primitive Earth — Atmosphere & Magnetic Reversals

- Primitive Earth ka atmosphere reducing tha: methane, ammonia, water vapour, carbon dioxide, carbon monoxide, nitrogen, hydrogen, neon etc. present the, lekin free oxygen absent tha. [Q2]
- Earth ka magnetic field history mein har kuch hundred-thousand years mein reverse hota raha; source ke mutabik last 5 million years mein at least 20 reversals hue. [Q3]
- Earth ke earliest atmosphere mein oxygen nahi tha; living organisms ke origin ke baad, especially oxygen-producing cyanobacteria jaise organisms ne atmosphere ko modify kiya. [Q3]

3. Origin of Life — Age of Life & Cyanobacteria

- Source ke Q4 ke options ke context mein life on Earth ka emergence approximately 2 billion years ago diya gaya hai; same explanation modern scientific consensus ko ~3.5–3.8 billion years ago batati hai. [Q4]
- Blue-green algae (Cyanobacteria) ko source oldest organism batata hai. Ye photosynthesis karte hain aur gaseous oxygen release karke Earth par biodiversity ke development mein important role nibhate hain. [Q5]



EXAM TRAP — Q4 — Option-based answer vs source note

- ▶ Question ka marked answer 2 billion years hai because wahi provided options mein fit hota hai. Source explanation khud modern consensus ko ~3.5–3.8 billion years ago note karti hai. [Q4]

4. Darwin — Origin of Species & Natural Selection

- Charles Darwin ne natural selection ko apni book 'The Origin of Species' mein explain kiya; source publication year 1859 deta hai aur ise evolutionary biology ki foundation batata hai. [Q6]
- Darwin ki natural selection theory ke main points: Overproduction, Struggle for existence, Variation, aur Survival of the fittest. [Q7]
- Widely accepted evolution theory based on natural selection ko source Darwin se associate karta hai. [Q8]
- Darwinian framework mein natural selection evolution ka main/basic mechanism diya gaya hai. [Q12]

5. Lamarckism — Acquired Characters & Use/Disuse

- French biologist Lamarck ko source life evolution ko formally public ke saamne explain karne wala first person batata hai. Uska 'Theory of Inheritance of Acquired Characters' 1801 mein present hua aur 1809 mein 'Philosophie Zoologique' mein publish hua. [Q9]
- Lamarckism ke context mein snakes ke limbs ka loss 'use and disuse of organs' se explain kiya gaya hai — burrowing ke dauran limbs ka use band hone se generations mein unka loss hua. [Q13]

6. Mutation Theory — Hugo de Vries & Genetic Variation

- Hugo de Vries ne mutation theory of evolution di. Source ke mutabik new species continuous variations ke bajay sudden variations (mutations) se form ho sakti hain; mutations heritable hoti hain. [Q10]
- Gene aur chromosomal mutations ko source biological changes ka ultimate source / evolution ka 'raw material' batata hai, jo large-scale diversity ko lead kar sakta hai. [Q14]
- New species ke origin ke liye source mutation ko most responsible factor batata hai because mutations new genetic variations introduce karti hain. [Q15]

7. Important Theory / Scientist Match

- Correct match: Theory of mutation → Hugo de Vries; Theory of evolution → Charles Darwin; One gene—one enzyme hypothesis → Beadle & Tatum; Operon concept → Jacob & Monod. [Q11]

EXAM TRAP — Darwin vs Lamarck vs De Vries

- ▶ Darwin → Natural selection / Origin of Species. [Q6, Q7, Q8, Q12]
- ▶ Lamarck → Inheritance of acquired characters / use and disuse; source says first formal explanation of evolution to public. [Q9, Q13]
- ▶ Hugo de Vries → Mutation theory. [Q10, Q11]

8. Human Evolution — Cro-Magnon & Denisovans

- Given options mein Cro-Magnon man ko modern human ka latest ancestor diya gaya hai. Source Cro-Magnon ko early modern humans ke liye common name batata hai; known remains ~43,000 years old aur brain capacity ~1,600 cc mention karta hai. [Q16]
- Denisovans / Denisova hominins ek extinct archaic early-human species/subspecies the jo Asia mein Lower aur Middle Paleolithic ke dauran rahe. First identification 2010 mein Siberian Denisova Cave se finger bone ke basis par hui. [Q17]

9. Archaeopteryx — Ancient Bird & Connecting Link

- Archaeopteryx ko source Jurassic age ka most ancient bird batata hai, approximately 150 million years old; ise oldest-known birds mein count kiya gaya hai. [Q18]
- Archaeopteryx Reptilia aur Aves ke beech connecting link hai: reptilian features mein teeth aur long bony tail, aur bird features mein feathers aur wings milte hain. [Q19]

10. Mammoth, Dinosaurs & Evolutionary Sequence

- Source ke question framing mein Mammoth ko elephant ka ancestor diya gaya hai. Explanation ke mutabik extinct mammoths modern African/Asian elephants ke closely related hain aur woolly mammoths + modern elephants ka common ancestor ~6 million years ago tha. [Q20]
- Dinosaurs Mesozoic reptiles the. Source Mesozoic ko reptiles/dinosaurs ka age batata hai, roughly 250–65 million years ago, aur isme Triassic, Jurassic aur Cretaceous periods include hote hain. [Q21]
- Era-based direct question mein dinosaurs ka correct era Mesozoic hai. [Q22]

- Evolutionary history mein amphibians reptiles se pehle aur reptiles mammals se pehle aaye. Source sequence: Salamander (amphibian, ~360 mya) → Python (reptile, ~300 mya) → Kangaroo (marsupial mammal, ~145 mya). [Q23]